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Fundamentals of Java

Q.1 Consider a directory structure created on the Window system A directory named AlphaProject has been created It contains two sub-directories named source and classes. The source directory contains a Java file named MyProgram.java Assume that the current directory has been set to AlphaProject Then, what will be the result of the following command invocation on the command prompt javac -d classes source/MyProgram.java	
A. If the command is executed successfully then MyProgram class file is added in the source directory	B. If the command is executed successfully then MyProgram class file is added in the current directory
C. If the command is executed successfully then MyProgram class file is added in the classes directory	D. The MyProgram java file is not compiled successfully because the command result in an error
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, Page no 39, Table 2.1	

Q.2 "Given that the root directory contains a subdirectory called "BankApp" that contains files for a Java application The following command is invoked from the root directory in order to create a JAR file containing the "BankApp" directory jar cf X Y. where X and Y are arguments. Which of the following statements correctly identify the arguments X and Y in the jar command?"	
A. X should be the file name of the JAR file. and Y should be "BankApp"	B. X should be "BankApp" and Y should be the file name of the JAR file
C. Both X and Y specifies file name of the JAR file	D. If a file name is not specified. then file named BankApp jar will be created
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Page no 331, Table 9.3	

Q.3 In functional programming, identify the symbol that is used to represent the method reference.	
A. :	B. ->
C. ::	D. -
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 524, Heading 15.9	

Q.4 Which of the following package contains the Stream API interfaces and classes?	
A. Java.xml.stream	B. Java.io
C. Javax.util.stream	D. Java.lang
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 540, Heading 16.1	

Q.5 Which of the following classes contains the method splitAsStream(CharSequence) used to generate streams?	
A. Stream	B. Pattern
C. Collection	D. BufferedReader
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, Page no 543, Heading 16.3	

Prepared By Abrar Ahmed

Q.6 Identify the default policy of @Retention annotation.	
A. RetentionPolicy.Runtime	B. RetentionPolicy.Class
C. RetentionPolicy.Source	D. RetentionPolicy.Value
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, Page no 499, Heading 14.5	

Q.7 Which of the following APIs can be used to access annotations on any type such as class, interface, or methods?	
A. Inherited	B. Documented
C. Repeatable	D. Reflection
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, Page no 5.6, Heading 14.8	

Q.8 Identify the statements that are correct with respect to mapping output of Simple Base64 encoding type.	
1. Outputs a set of characters between A-Z and a-z	
2. Encoder adds line feed at the end of the output	
3. Decoder accepts only A-Z,a-z,0-9, and +	
A. Only 1	B. 2 and 3
C. Only 2	D. 1 and 3
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, Page no 508, Heading 14.10	

Q.9 Identify the statements that are true about a default method.	
A) Reduces the gaps between interfaces and abstract classes	
B) Eliminates need for utility class	
C) Assists in base implementation class removal	
D) Extends interfaces after breaking implementation classes	
A. A and B	B. B and C
C. A, B, and C	D. All of these
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Page no 529, Heading 15.11	

Q.10 The _____ keyword is used in an instance method or a constructor to refer to the current object	
A. super	B. static
C. this	D. new
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 244, Heading 7.4.4	

Q.11 Which of the following methods of an Exception class return the details about the exception that has occurred?	
A. getMessage()	B. getCause()
C. printStackTrace()	D. StackTraceElement()
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Page no 431, Table 12.4	

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Q.12 Given the statements: SavingAccount objSA = new SavingAccount(); Account objAcc = objSA; Which of the following options correctly identifies the type of casting performed in these statements?	
A. Implicit	B. Explicit
C. Upcasting	D. Downcasting
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 362, Heading 10.3.2	

Q.13 Consider the following declaration of a StringBuilder class: StringBuilder objSb = new StringBuilder("ABC"); Which of the following methods of the StringBuilder class when used will display the output as ABC123?	
A. objSb.concat['123'];	B. objSb.insert(1,"123");
C. objSb.insertlast['123'];	D. objSb.append['123'];
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, Page no 284, Heading 8.3.4	

Q.14 Identify the method that is used to compare two String objects?	
A. Comparator()	B. Comparable()
C. Compare()	D. compareTo()
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, Page no 280, Heading 8.3.2	

Q.15 Consider the following code snippet: 1. Iterator itr = arraylist.iterator(); 2. // insert code here 3. { 4. System.out.println(itr.next()); 5. } Which of the following statement added at line 2 will compile the code successfully?	
A. while(itr.moreElements())	B. while(itr.hasMoreElements())
C. while(itr.hasNext())	D. while(itr.next())
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 275, Chapter 8, Code Snippet 8	

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Q.16 Which of the following keywords are used to explicitly raise an exception from a method? (Choose two correct answers)	
A. try	B. catch
C. throw	D. throws
Correct Answer: C,D	
Reference: Book name: Fundamental programming in Java, Page no 453, Chapter 12 Summary	

Q.17 Which of the following command is used to extract contents of a jar file	
A. jar cfjar-file-name	B. jar xfjar-file-name
C. jar tjar-file-name	D. java -jar jar-file-name
Correct Answer: B	
Reference: Book name: Fundamental programming in Java , Page no 331, Table 9.3	

Q.18 Which of the following statements are true regarding transient modifier? (Choose three correct answers)	
A. Can only be used with instance variables	B. Useful to prevent security sensitive data from being copied to a source in which no security mechanism has been implemented
C. informs the JVM not to store the variable when the object, in which it is declared, is serialized	D. Involves performance issues and also increases cost
Correct Answer: A,C	
Reference: Book name: Fundamental programming in Java, Page no 310, Heading 9.2.3	

Q.19 Which of the following statements are true regarding local inner classes? (choose two correct answers)	
A. Associated with an instance of the enclosing class	B. Cannot access private members of the enclosing class
C. Can access any local variables, method parameters, or exception parameters with scope of the local method declared as final	D. Can implement anonymous interfaces
Correct Answer: A,C	
Reference: Book name: Fundamental programming in Java, Page no 401, Heading 11.3.3	

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Q.20 Consider the following method declarations? <pre> Public short getInt(){ Int s=5; Return s; } </pre> Which of the following statements is true regarding the method declaration?	
A. The method parameter list cannot be empty	B. The method will not compile as the variable s need to cast
C. The method will compile successfully	D. The statement int s=5 throws runtime exception
Correct Answer: B	
Reference: Book name: Fundamental programming in Java , No direct reference available	

Q.21 The _____ class is declared without a name within a code block such as a body of a method.	
A. Local inner class	B. Anonymous class
C. Static Nested Class	D. Final class
Correct Answer: B	
Reference: Book name: Fundamental programming in Java ,Page no 405, Heading 11.3.4	

Q.22 Which of the following interface does not inherit from the collection Interface?	
A. Set	B. List
C. Map	D. Queue
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 271, Fig 8.8	

Q.23 What is the defined offset from the starting point for one instant of the instant class?	
A. January 1, 1900-00:00-UTC	B. January 1, 1970-00:00-GMT
C. January 1, 1900-00:00-GMT	D. January 1, 1970-00:00-UTC
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, Page no 462, Heading 13.2.2	

Q.24 The syntax for @Retention annotation is _____	
A. @Retention	B. @Retention(value)
C. @Retention(RetentionPolicy value)	D. @Retention()
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, Page no 499, code snippet 9	

Q.25 Identify the shortest manner in which you can declare a java annotation	
A. Prefix @	B. Suffix @
C. Prefix #	D. Suffix #
Correct Answer: A	

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Reference: Book name: Fundamental Programming in Java, Page no 492,Heading14.3

Q.26 Which of the following APIs represents a date without time -zone information?

- | | |
|--------------|------------------|
| A. Period | B. Year |
| C. LocalDate | D. ZonedDateTime |

Correct Answer: C

Reference: Book name: Fundamental Programming in Java, Page no 463,Heading 13.2.3

Q.27 In Java 8, which of the following classes is used in place of TimeZone.getDefault() method?

- | | |
|------------|-------------|
| A. System | B. Clock |
| C. Instant | D. Duration |

Correct Answer: B

Reference: Book name: Fundamental Programming in Java, Page no 459,Heading 13.2.1

Q.28 Consider a software app that must be developed analyzing the interest of different zonal people in the UTC time with respect to the time –zone of respective countries. Which of the following classes in Date-Time API will you use to signify the exact time of day without any time-zone data?

- | | |
|--------------|------------|
| A. LocalDate | B. Instant |
| C. LocalTime | D. Enum |

Correct Answer: C

Reference: Book name: Fundamental Programming in Java, Page no 467, Heading 13.2.5

Q.29 Which of the following functions on any ZoneId ensures that a fixed offset ID will be formed as ZoneOffset?

- | | |
|---------------|-----------------|
| A. getValue() | B. now() |
| C. of() | D. normalized() |

Correct Answer: D

Reference: Book name: Fundamental Programming in Java, Page no 477, Heading 13.2.13

Q.30 The _____ commands is used to interpret and run the java bytecode

- | | |
|----------|------------|
| A. javac | B. java |
| C. javap | D. javadoc |

Correct Answer: B

Reference: Book name: Fundamental Programming in Java, page no 40, Below Fig 2.6

Q.31 Which of the following options are the correct implementations of a Queue? (Choose two correct answer)

- | | |
|-------------------|-------------------|
| A. Linked List | B. Array List |
| C. Priority Queue | D. Circular Queue |

Correct Answer: A, C

Reference: Book name: Fundamental Programming in Java, page no 271, Below table 8.1

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Q.32 Consider the following code:

```
public class TestArray{
    public static void main(String [] args) {
        int arr[] = {1,2,3,4};
        for(int x = 1; x < arr.length; x++) {
            System.out.print(arr[x++] + " ");
        }
    }
}
```

What is the result of the code on execution?

A. 1 3

B. 2 3 4

C. 1 2 3 4

D. 2 4

Correct Answer: D

Reference: Book name: Fundamental Programming in Java, Reference not available

Q.33 Match the following modifiers with the description

SI NO	Left Column	SI NO	Right Column	Correct Answer for Column A			
A1	Volatile	B1	It is used to invoke a method that resides outside JVM	B1	B2	B3	B4
A2	Native	B2	It is used with a variable that will not be stored as part of object in the persistence storage	B1	B2	B3	B4
A3	Transient	B3	It is used when the modification of a class or data member is to be restricted	B1	B2	B3	B4
A4	Final	B4	It allows the content of a variable to be synchronized across all running threads	B1	B2	B3	B4

Reference: Book name: Fundamental Programming in Java, page no 308 – 311, Heading 9.2.1 – 9.2.4

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Q.34 Consider the following code:

```
Public class Test{
    Static int count = 0;
    Test() { while(count < 10) new Test(++ count); }
    Test(int x) {}
    Public static void main(String[] args) {
        new Test();
        new Test(count);
        System.out.println(count++);
    }
}
```

What is the output of the code on execution?

A. 10	B. 11
C. 12	D. Compilation fails
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Reference not available	

Q.35 Consider the following declarations: (choose two correct answers)

```
boolean b = false;
int i = 7;
double d = 1.23;
float f = 4.56f;
```

Which of the following options are the correct format specifiers to format and print the value of these variables on the console?

A. System.out.printf(“ %b”, b);	B. System.out.format(“ %d”, i);
C. System.out.format(“ %d”, d);	D. System.out.printf(“ %i”, i);
Correct Answer: A, B	
Reference: Book name: Fundamental Programming in Java, Reference not available	

Q.36 plusSeconds() method adds seconds in a calculation belongs to the _____ class.

A. Duration	B. Period
C. Instant	D. Year
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, page no 460	

Q.37 In how many ways can the Math.abs() method be overloaded?

A. 1	B. 2
C. 3	D. 4
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, page no 591, Heading 18.7.1	

Prepared By Abrar Ahmed

Q.38 Nashorn uses the _____ API of JVM specification.	
A. Reflection	B. ClassFilter
C. InvokeDynamic	D. Streams
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, page no 584, Heading 18.1	

Q.39 Which of the following methods are short-circuiting terminal operations?	
a. forEach b. anyMatch c. findAny d. max*	
A. a and b	B. b and c
C. a,b,c	D. All of these
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, page no 546, Heading 16.6.3	

Q.40 Identify the method that does not start its process until a terminal operation is called.	
A. limit()	B. map()
C. filter()	D. sort()
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, page no 547, Heading 16.8.2	

Q.41 identify the line of code that prints the text Hello World as output.	
A. ()-> System.out.println("Hello World");	B. (string)-> System.out.println("Hello World");
C. ()-> System.out.println "Hello World";	D. (string)-> System.out.println "Hello World";
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Reference not available	

Q.42 Which method of the Function interface is comparable to and Then() Function is reversed sequence?	
A. Identity	B. Apply
C. Compose	D. Default
Correct Answer: C	
Reference: Book name: Fundamental Programming in Java, page no 569	

Q.43 Consider the following code snippet:	
<pre> ArrayList Marks = new ArrayList(); Marks.add(67); Marks.add(90); Marks.add(78); Marks.add(90); Marks.add(34); Marks.add(56); Collections.sort(marks); System.out.println(marks); </pre>	

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What is the output of the code?	
A. [34, 56, 67, 78, 90,90]	B. [90, 90, 78, 67, 56, 34]
C. [90, 78, 67, 56, 34]	D. Compilation error
Correct Answer: A	
Reference: Book name: Fundamental Programming in Java, Page no 274, Heading 8.2.5	

Q.44 which of the following statements truly specifies the characteristics of static methods in Java? (Choose two correct answers).	
A. Can access only static data	B. Can invoke static and normal methods
C. Can use super keywords	D. Cannot use this keyword
Correct Answer: A, D	
Reference: Book name: Fundamental Programming in Java, page no 314, Heading 9.3.2	

Q.45 A stream containing _____ intermediate operators has no limitations.	
A. 5	B. 10
C. 15	D. n
Correct Answer: D	
Reference: Book name: Fundamental Programming in Java, page no 544, Heading 16.6.1	

Q.46 Default method cannot override a method from which of the following classes?	
A. Java.lang.class	B. Java.util.collections
C. Java.util.stream	D. Java.lang.object
Correct Answer:	
Reference: Book name: Fundamental Programming in Java, page no 529, Heading 15.11	

Q.47 Consider the following declaration: double d = 15.3446; Which of the following options shows the correct statement that produces the output 015.345 ?	
A. System.out.printf(" %07d \n",d);	B. System.out.printf(" %07.3f \n",d);
C. System.out.printf(" %07.3d \n",d);	D. System.out.printf(" %07.3e \n",d);
Correct Answer: B	
Reference: Book name: Fundamental Programming in Java, page no 87, Heading 3.7.2	

Q.48 Which of the following statements are true regarding the Java Virtual Machine? (Choose two correct answers)	
A. It is an execution engine which creates a platform independent execution environment for executing compiled code	B. The same implementation of JVM is available for different platforms
C. The same bytecode can be executed by the same implementation of JVM	D. It is an imitation of a Java processor on the physical machine
Correct Answer: A,D	
Reference: Book name: Fundamental Programming in Java, page no 14	

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Q.49 Identify which among the following are the valid development tools provided by Java Development Kit. [Choose two correct answers]	
A. javah	B. docjava
C. javadoc	D. jar
Correct Answer: C,D	
Reference: Book name: Fundamental Programming in Java, page no 17	